

WEB DEVELOPMENT

Viva Questions & Answers

SPPU | SE Computer Engineering | VSE-270-COM | 2024 Pattern

60 Questions | General Web | HTML | CSS | JavaScript | PHP

General Web

7 Questions

HTML

10 Questions

CSS

10 Questions

JavaScript

11 Questions

PHP

12 Questions

GENERAL WEB

Q
1

General
I

What is a web browser?

Ans: Software that lets you view websites. Sends a request to a server and shows the HTML/CSS/JS response. Examples: Chrome, Firefox, Edge.

Q
2

General
I

What is HTTP and HTTPS?

Ans: HTTP = HyperText Transfer Protocol. Rules for sending data between browser and server. HTTPS = Secure version where data is encrypted. Always use HTTPS for safety.

Q
3

General
I

What is a web server and web client?

Ans: Client = your browser (sends requests). Server = computer that stores the website and sends back a response. Example: You open Google, your browser (client) asks Google's server for the page.

Q
4

Genera
I

Difference between static and dynamic web pages?

Ans: Static: Fixed content, same for everyone, uses only HTML/CSS. Dynamic: Content changes based on user/time/data, uses PHP or JavaScript. Example: Facebook is dynamic.

Q
5

Genera
I

What is a URL?

Ans: URL = Uniform Resource Locator. The address of a web page. Example: <https://www.google.com>. Parts: protocol (https), domain (google.com), path (/search).

Q
6

Genera
I

Difference between frontend and backend?

Ans: Frontend: What the user sees in the browser, HTML, CSS, JavaScript. Backend: Server-side logic and databases, PHP, Python, Node.js.

Q
7

Genera
I

What is DOM?

Ans: DOM = Document Object Model. A tree-like structure the browser creates from HTML. JavaScript uses it to find, change, add, or delete page elements without reloading the page.

HTML

Q
8

HTML

What is HTML? What does it stand for?

Ans: HTML = HyperText Markup Language. It is the structure/skeleton of a webpage. We use tags like h1, p, form, table to add and organize content on the page.

Q
9

HTML

Difference between div and span?

Ans: div: Block-level element, takes full width, starts on a new line. Used for sections and layouts. span: Inline element, stays in the same line. Used for styling part of text.

Q
1
0

HTML

Difference between block and inline elements?

Ans: Block elements take full width and start on a new line. Examples: h1, p, div, table. Inline elements stay on the same line. Examples: span, a, b, img.

Q
1
1

HTML

What are semantic HTML tags?

Ans: Tags that describe their meaning clearly. Examples: header, nav, section, article, footer. They improve readability, make code clean, and help SEO (search engine optimization).

Q
1
2

HTML

What is the form tag? What are method and action?

Ans: The form tag collects user input. action = URL where data is sent. method = how data is sent: GET shows data in URL, POST hides it. Example: form method=POST action=page.php

Q
1
3

HTML

Difference between GET and POST methods?

Ans: GET: Data is visible in the URL, not secure, used for searching. POST: Data is hidden in the request body, secure, used for login, registration, and form submissions.

Q
1
4

HTML

What are different input types in HTML?

Ans: text, email, password, number, date, radio, checkbox, file, submit, button, hidden. Example: input type=email automatically checks for valid email format.

Q
1
5

HTML

Difference between id and class in HTML?

Ans: id: Unique, used on one element only. In CSS use #myid, in JS use getElementById. class: Can be used on many elements. In CSS use .myclass. id has higher priority than class.

Q
1
6

HTML

What is the alt attribute in the img tag?

Ans: alt = alternative text shown when image fails to load. Also helps screen readers for blind users. Example: img src=photo.jpg alt=My Photo. It is a good practice to always add alt.

Q
1
7

HTML

What is a hyperlink? How to create it?

Ans: A clickable link that goes to another page. Syntax: a href=URL then Link Text then /a. To open in new tab add target=_blank. href stands for Hypertext REference.

CSS

Q
1
8

CSS

What is CSS? What does it stand for?

Ans: CSS = Cascading Style Sheets. Used to style HTML elements, change color, font, size, spacing, and layout of a webpage.

Q
1
9

CSS

What are the 3 types of CSS?

Ans: 1. Inline: style attribute on the element like style=color:red. 2. Internal: style tag inside head section. 3. External: Separate .css file linked with link rel=stylesheet href=style.css.

Q
2
0

CSS

What is the CSS Box Model?

Ans: Every HTML element is a box with 4 layers from inside out: Content, then Padding (space inside border), then Border, then Margin (space outside element). Used to control size and spacing.

Q
2
1

CSS

Difference between margin and padding?

Ans: Padding: Space between the content and the border, inside the element. Margin: Space outside the border, the gap between elements on the page.

Q
2
2

CSS

What is CSS Grid? When is it used?

Ans: CSS Grid is a 2D layout system that handles rows AND columns. Use: display:grid with grid-template-columns: repeat(3,1fr). Used for card layouts, restaurant menus, dashboards.

Q
2
3

CSS

What is a CSS selector? Name different types.

Ans: A selector targets HTML elements to apply styles. Types: element (p), class (.box), id (#header), universal (*), descendant (div p), pseudo-class (a:hover).

Q
2
4

CSS

What is Bootstrap?

Ans: Bootstrap is a free CSS framework by Twitter. It provides ready-made classes for grids, buttons, forms, navbars. Makes websites responsive (mobile-friendly) without writing much CSS.

Q
2
5

CSS

How does Bootstrap Grid work?

Ans: Bootstrap divides the screen into 12 columns. Use col-md-4 which takes 4 out of 12 = one-third width on medium screens. Always wrap columns in a row div inside a container div.

Q
2
6

CSS

What is a responsive website?

Ans: A website that adjusts its layout based on screen size. Looks good on mobile, tablet, and desktop. Achieved using Bootstrap grid, flexbox, or CSS media queries.

Q
2
7

CSS

What is display flex used for?

Ans: Flexbox arranges elements in a row or column flexibly. Useful for centering elements and equal spacing. Use justify-content and align-items to control alignment of items.

JAVASCRIPT

Q
2
8

JavaSc
ript

What is JavaScript? Why is it used?

Ans: JavaScript is a programming language that runs in the browser. Used to make web pages interactive: form validation, dynamic content, DOM manipulation, calculators, quizzes, etc.

Q
2
9

JavaSc
ript

Difference between var, let, and const?

Ans: var: Old way, function scoped, can be redeclared. let: Block scoped, value can be changed but not redeclared. const: Block scoped, value cannot be changed after declaration.

Q
3
0

JavaSc
ript

What is a function in JavaScript?

Ans: A function is a reusable block of code. Define: `function add(a,b){ return a+b; }` Call it: `add(2,3)` returns 5. Functions avoid writing the same code again and again.

Q
3
1

JavaSc
ript

What is the switch-case statement?

Ans: Used when you have multiple conditions to check for the same variable. Each case matches one value. Always add break after each case to stop. Use default for when no case matches.

Q
3
2

JavaSc
ript

What is DOM manipulation?

Ans: Using JavaScript to find and change HTML elements after the page loads. Key methods: `getElementById`, `querySelector`, `innerHTML`, `innerText`, `createElement`, `appendChild`.

Q
3
3

JavaSc
ript

Difference between innerHTML and innerText?

Ans: `innerHTML`: Gets or sets HTML content including tags. `el.innerHTML = '<b>Hello</b>'` shows bold text. `innerText`: Gets or sets only plain text, HTML tags are not processed.

Q
3
4

JavaSc
ript

How do you validate a form using JavaScript?

Ans: Get the input value using `getElementById`, check if it is empty or use regex to match a pattern. If invalid, show an error message and return false to stop form submission.

Q
3
5

JavaSc
ript

What is a regex? Give example for email.

Ans: Regex = Regular Expression, a pattern to match text. Email regex: `/^[^s@]+@[^s@]+\.[^s@]+$/`. Use `regex.test(value)` which returns true if the value matches the pattern.

Q
3
6

JavaSc
ript

What is an event in JavaScript?

Ans: An event is an action the user performs. Examples: `onclick` (button click), `onsubmit` (form submit), `onchange` (dropdown change), `onkeyup` (typing). Add `onclick=myFunction()` in HTML.

Q
3
7

JavaSc
ript

What does `parseFloat` and `parseInt` do?

Ans: Both convert a string to a number. `parseInt('42')` gives 42 (whole number). `parseFloat('3.14')` gives 3.14 (decimal). Input values are always strings so conversion is needed for math.

Q
3
8

JavaSc
ript

What is `createElement` and `appendChild`?

Ans: `createElement('p')` creates a new p element in memory. `appendChild(element)` adds it inside a parent element on the page. Together they add dynamic content without reloading.

Q
3
9

PHP

What is PHP? What does it stand for?

Ans: PHP = PHP: Hypertext Preprocessor. A server-side scripting language. PHP code runs on the server and sends plain HTML to the browser. Used for dynamic websites, forms, and databases.

Q
4
0

PHP

Difference between PHP and JavaScript?

Ans: JavaScript runs in the browser (client-side). PHP runs on the server (server-side). PHP processes data on the server before sending HTML. JS makes the page interactive after loading.

Q
4
1

PHP

How do you start and end PHP code?

Ans: PHP starts with `<?php` and ends with `?>`. Short echo tag: `<?=$variable ?>` prints a value directly. PHP code must be in a .php file and run on a server like XAMPP or WAMP.

Q
4
2

PHP

What is \$_POST and \$_GET in PHP?

Ans: Superglobals (special arrays) that hold form data. `$_POST['name']` gets data sent via POST method (hidden). `$_GET['name']` gets data from the URL (visible). Use POST for sensitive data.

Q
4
3

PHP

How to check if a form is submitted in PHP?

Ans: Use: `if ($_SERVER['REQUEST_METHOD'] == 'POST') { process data here }`. This checks if the page was submitted via a form, not just opened. Then safely access `$_POST` values inside.

Q
4
4

PHP

What is the date() function in PHP?

Ans: Returns current date and time. date('d-m-Y') gives 29-04-2025. date('h:i:s A') gives 10:30:00 AM. date('l') gives Wednesday. date('Y') gives just the year number.

Q
4
5

PHP

Name important PHP string functions.

Ans: strlen() = length | strrev() = reverse | strtoupper() = uppercase | strtolower() = lowercase | ucfirst() = first letter capital | ucwords() = each word capital | substr(\$s,0,5) = substring | str_replace() = replace text | ltrim(\$s,'0') = remove leading zeros

Q
4
6

PHP

Name important PHP array functions.

Ans: array_sum() = total of all values | max() = highest value | min() = lowest value | count() = number of elements | in_array(\$val,\$arr) = check if value exists | array_search() = find index | implode(',', \$arr) = join array to a string

Q
4
7

PHP

Difference between echo and print in PHP?

Ans: Both display output on screen. echo is faster and can print multiple values separated by comma. print returns 1 so it can be used in expressions. In practice always use echo.

Q
4
8

PHP

How do you create an array in PHP?

Ans: \$marks = array(85, 72, 90); or short form: \$marks = [85, 72, 90]; Access using index: \$marks[0] gives 85. Named array: \$s = array('name'=>'Rahul', 'age'=>20); Access: \$s['name'].

Q
4
9

PHP

What is in_array() function?

Ans: in_array(\$value, \$array) checks if a value exists in an array. Returns true if found, false if not.
Example: in_array(103, \$rolls) is used in the roll number search program.

Q
5
0

PHP

What is htmlspecialchars() and why use it?

Ans: Converts special characters like < > & to safe HTML codes. Used to prevent XSS (Cross-Site Scripting) attacks when showing user input on a page. Always use it with \$_POST data.

QUICK TIPS FOR TOMORROW

- Most Asked:** What is the difference between GET and POST?
- Most Asked:** What are the 3 types of CSS? (Inline, Internal, External)
- Most Asked:** What is DOM? What is getElementById?
- Most Asked:** What is PHP and how is it different from JavaScript?
- Most Asked:** What is the difference between static and dynamic webpage?
- Remember:** PHP runs on SERVER | JavaScript runs in BROWSER
- Remember:** POST = hidden/secure | GET = visible in URL
- Remember:** Inline CSS > Internal CSS > External CSS (priority order)

All the best for your exam tomorrow!